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# Diabetes risk prediction model based on community follow-up data using machine learning

Liangjun Jiang [a], Zhenhua Xia [b], Ronghui Zhu [c], Haimei Gong [a], Jing Wang [d], Juan Li [e], Lei Wang [a,*]

[a] College of Information and Communication Engineering, State Key Lab of Marine Resource Utilisation in South China Sea, Hainan University, Haikou, China
[b] Electronics & Information School of Yangtze University, Jingzhou, China
[c] Shenzhen Nanshan Medical Group HQ, Shenzhen, China
[d] E-link Wisdom Co., Ltd, Shenzhen, China
[e] Haizhu District Community Health Development Guidance Center, Guangzhou, China

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ABSTRACT

Diabetes is a chronic metabolic disease characterized by hyperglycemia, the follow-up management of diabetes patients is mostly in the community, but the relationship between key lifestyle indicators in community follow-up and the risk of diabetes is unclear. In order to explore the association between key life characteristic indicators of community follow-up and the risk of diabetes, 252,176 follow-up records of people with diabetes patients from 2016 to 2023 were obtained from Haizhu District, Guangzhou. According to the follow-up data, the key life characteristic indicators that affect diabetes are determined, and the optimal feature subset is obtained through feature selection technology to accurately assess the risk of diabetes. A diabetes risk assessment model based on a random forest classifier was designed, which used optimal feature parameter selection and algorithm model comparison, with an accuracy of 91.24% and an AUC corresponding to the ROC curve of 97%. In order to improve the applicability of the model in clinical and real life, a diabetes risk score card was designed and tested using the original data, the accuracy was 95.15%, and the model reliability was high. The diabetes risk prediction model based on community follow-up big data mining can be used for large-scale risk screening and early warning by community doctors based on patient follow-up data, further promoting diabetes prevention and control strategies, and can also be used for wearable devices or intelligent biosensors for individual patient self examination, in order to improve lifestyle and reduce risk factor levels.

## 1. Introduction

Diabetes mellitus is a metabolic disorder characterized by hyperglycemia (Olisah et al., 2022; Khan et al., 2019). Type 2 diabetes (T2D) is an aggressive disease (Cioana et al., 2022). Early comorbidities and complications include hypertension, renal disease, polycystic ovary syndrome, and dyslipidemia (Cioana et al., 2023; Pinhas-Hamiel and Zeitler, 2007; Dabelea et al., 2017; Cioana et al., 2021; Cioana et al., 2022). It is a global public health problem that places a huge economic burden on health systems (Liu et al., 2023; Khan et al., 2019). The World Health Organization (WHO) has defined high-risk groups for diabetes (Han et al., 2018). The WHO report indicates that diabetes causes 1.5 million deaths worldwide each year (Bhat et al., 2022). Therefore, there is a need for more measures to enhance public awareness of the hazards of diabetes and improve the ability to predict the risk of diabetes. Establishing a diabetes risk prediction model and popularizing it among the population is an effective preventive measure for diabetes, which helps enhance the public's awareness of the risk of diabetes. Diabetes has become a major focus of research in machine learning and disease detection model design (García-Ordás et al., 2021; De Silva et al., 2020; Asgari et al., 2021; Yifei et al., 2023; Wei et al., 2022; McGurnaghan et al., 2021; Gautier et al., 2021; Kalagotla et al., 2021; Delshi Howsalya Devi et al., 2020; Lu et al., 2022; Hossain et al., 2021).

Previous studies have also shown that using physical examination data to build a disease risk assessment model has the potential to provide clinical guidance and early mass screening (Fadini et al., 2020).

* Corresponding author.
*E-mail addresses:* jiangliangjun@hainanu.edu.cn (L. Jiang), 100855@yangtzeu.edu.cn (Z. Xia), 261802759@qq.com (R. Zhu), ghm19830607@163.com (H. Gong), leyou_youle@163.com (J. Wang), 624821019@qq.com (J. Li), wanglei@hainanu.edu.cn (L. Wang).

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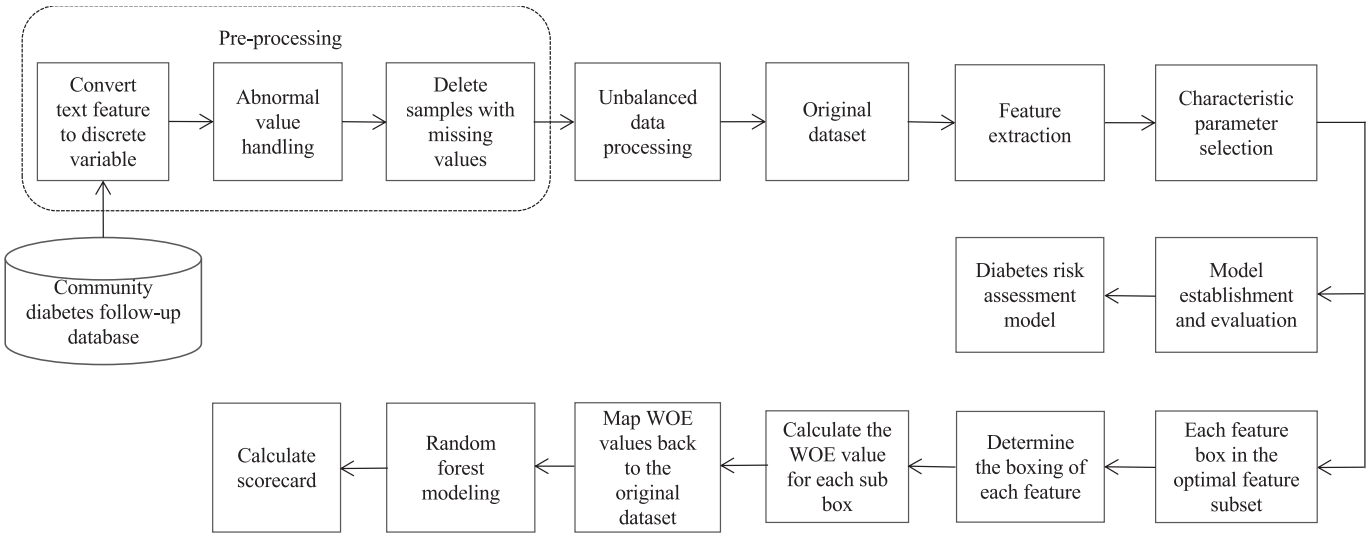



**Fig. 1.** Flowchart of building a diabetes risk prediction model based on machine learning community follow-up data.

Understanding an individual's glycemic risk can also guide the prevention and treatment of complications (Kavakiotis et al., 2017). Previous large-scale studies have shown that dietary and lifestyle changes can prevent or delay the development of diabetes in high-risk populations (Moon et al., 2021; Zheng et al., 2018). Some studies have conducted studies by bringing in laboratory data and trying to find characteristics that better guide patients' lifestyle. Hang Lai et al. (Lai et al., 2019) analyzed Canadian laboratory data through machine learning, and concluded that Fasting blood glucose, body mass index, high-density lipoprotein, and triglycerides are the best predictive parameters for diabetes, the GBM and logistic regression models performed better than the random forest and decision tree models. The ROC of the GBM model was 84.7% and the sensitivity was 71.6%. Liu, S., Gao, Y., Shen, Y. et al. (Liu et al., 2019) used BP neural network to model the five parameters of triglyceride, age, hypertension, alcohol consumption, and total cholesterol and obtained the highest prediction accuracy of 91.2%, the corresponding areas under curve (AUCs) is 0.78. Salliah et al. (Bhat et al., 2022) conducted learning and training of 6 machine algorithms based on clinical data sets collected from the Bandipora region of India from April 2021 to February 2022. Experiments showed that random forest had the highest accuracy rate of 98%, followed by multi-layer perceptron (90.99 %), Support Vector Machines (92%), Gradient Boosting (97%), Decision Trees (96%), and Logistic Regression (69%). This is similar to the findings of heart disease risk prediction studies (Ansari et al., 2023). Shahadat et al. (Uddin et al., 2022) considered 31 disease-related variables and conducted an analysis based on the guidelines of the Elixhauser Comorbidity Index. They developed a predictive model for type 2 diabetes using administrative claims data. The results showed that the random forest algorithm achieved the highest accuracy rate (85.06%), and age and comorbidities were identified as the most important predictive factors.

However, there are still several problems in the current research on diabetes models. First, the amount of training sample data is small, and the model does not have generalization ability. For example, the Pima Indian dataset (Bhat et al., 2022) only includes 768 data samples. Second, the performance of these models is unsatisfactory due to under-characterized laboratory data. In addition, diabetes management based on public health is more in community hospitals. Most of the data used in these samples are not directly from the community, and cannot be effectively used for large-scale screening and early warning in the community. To overcome the shortcomings of previous models, we developed a new model for assessing diabetes risk. The characteristic variables used in the model come from the daily follow-up data of community doctors in carrying out basic public health services. The dataset contains a large amount of data, including 252,176 follow-up records of people with diabetes in Haizhu District, Guangzhou City, from 2016 to 2023. The characteristics of the data in each region are unified, which is conducive to the large-scale popularization of the model. At the same time, a diabetes risk score card was designed to facilitate the rapid identification of patients by community follow-up doctors and the self-screening of patients.

**Table 1**
Overview and Description of Features of the Dataset.

| | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|
| BMI[a] | 23.95 | 2.80 | 12.65 | 22.06 | 23.6 | 25.47 | 39.84 |
| Age[b] | 65.46 | 11.97 | 18 | 57 | 65 | 75 | 108 |
| Smoking[c] | 0.94 | 3.77 | 0 | 0 | 0 | 0 | 20 |
| Staple_food[d] | 330.34 | 114.07 | 0 | 250 | 300 | 400 | 1300 |
| Exercise_frequency[e] | 3.78 | 1.73 | 0 | 3 | 3 | 5 | 14 |
| Exercise_time[f] | 34.19 | 14.25 | 0 | 30 | 30 | 40 | 100 |
| Drinking[g] | 0.12 | 0.71 | 0 | 0 | 0 | 0 | 25 |
| Systolic_BP[h] | 127.54 | 9.01 | 72 | 122 | 128 | 132 | 200 |
| Diastolic_BP[i] | 76.92 | 6.64 | 38 | 72 | 77 | 81 | 140 |
| Diabetes[j] | 0.22 | 0.42 | 0 | 0 | 0 | 0 | 1 |

(a) BMI: Body Mass Index. (b) Age: Age. (c)Smoking: Smoking status, measured in cigarettes, with 20 cigarettes per pack. (d) Staple_food: Staple food, measured in grams. (e) Exercise_frequency: Exercise frequency, the number of times exercising per week. (f) Exercise_time: Duration of a single exercise session, measured in minutes. (g) Drinking: Drinking status, measured in units of Chinese liquor. 500 g of beer = 1 unit of Chinese liquor, 4 units of red wine = 1 unit of Chinese liquor, 5 units of yellow wine = 1 unit of Chinese liquor. (h) Systolic_BP: Systolic blood pressure. (i) Diastolic_BP: Diastolic blood pressure. (j) Diabetes: Class variable (either 0 or 1). 0 means no diabetes, and 1 means having diabetes. In the dataset, there are 188,753 instances with label 0 and 63,423 instances with label 1.

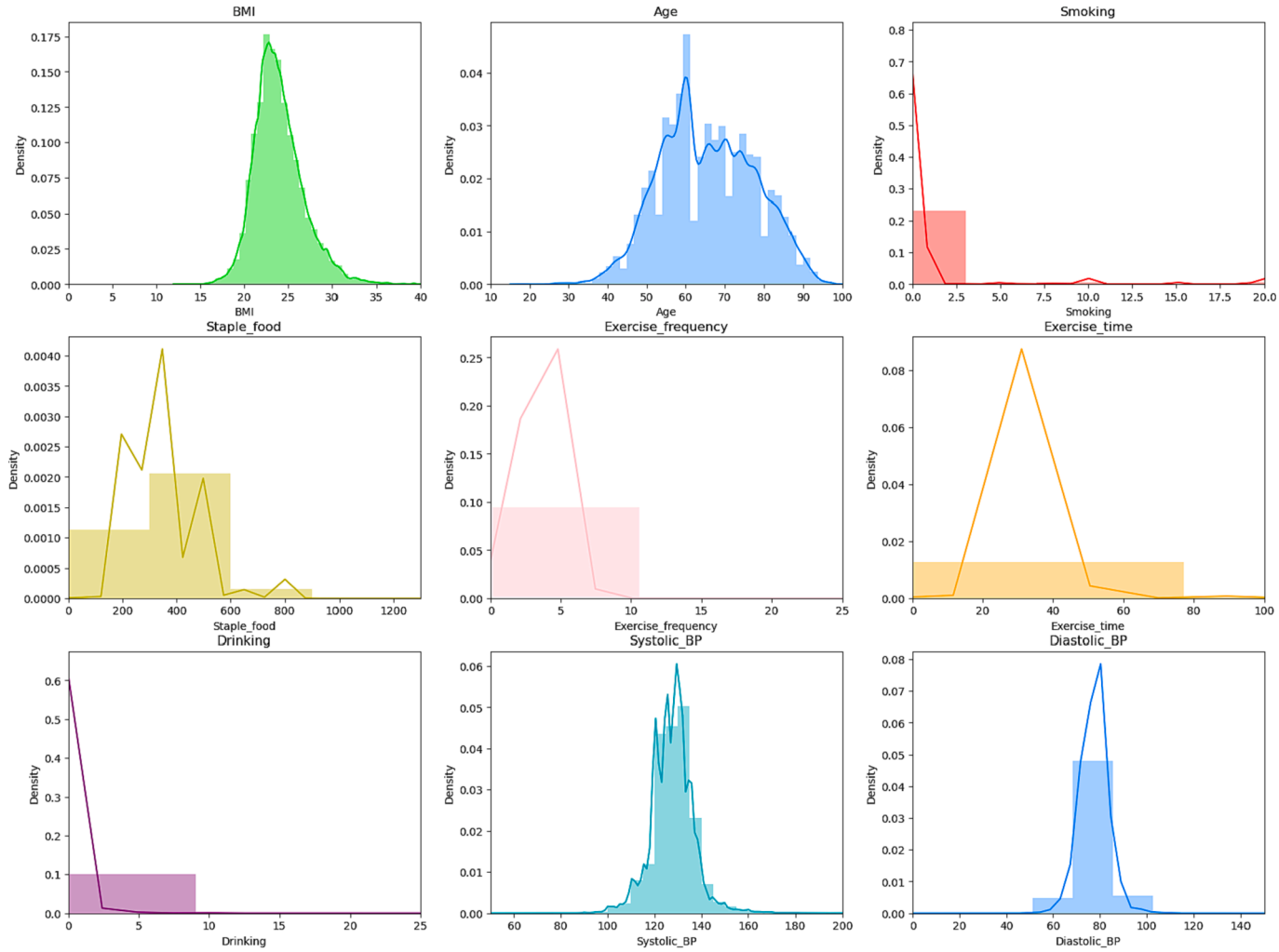
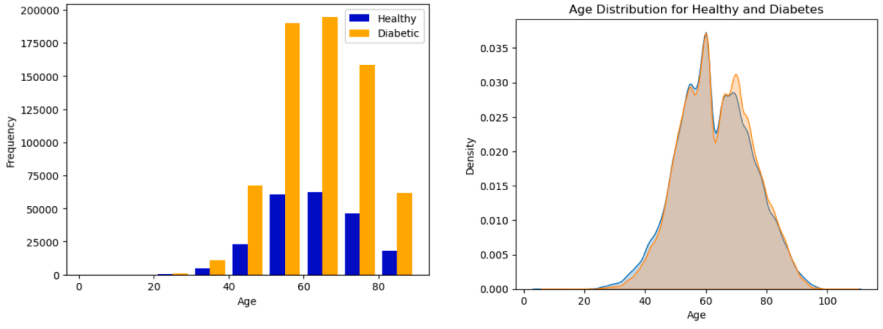



**Fig. 2.** The numerical distribution of the nine characteristic variables of the data set in this study.

## 2. Materials and methods

### 2.1. Study design

This study consists of many steps (Fig. 1). The first step is to preprocess the acquired community follow-up data (variable conversion, outlier processing, missing value processing) to obtain the research data set we use. The second step is to balance the research data set through oversampling, and select the optimal feature set through feature selection technology. The third step is to split the dataset selected based on the optimal features into a training dataset subset (containing 70% of the entire dataset) and a testing dataset subset (containing 30% of the entire dataset), and use four algorithm models for training to obtain the optimal prediction algorithm model. The fourth step is to establish a risk assessment card.




(a) (b)

**Fig. 3.** Age distribution of healthy individuals (blue) and people with diabetes (yellow). (a) The age distribution is displayed using a bar chart; (b) The age distribution is displayed using a kernel density estimation plot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

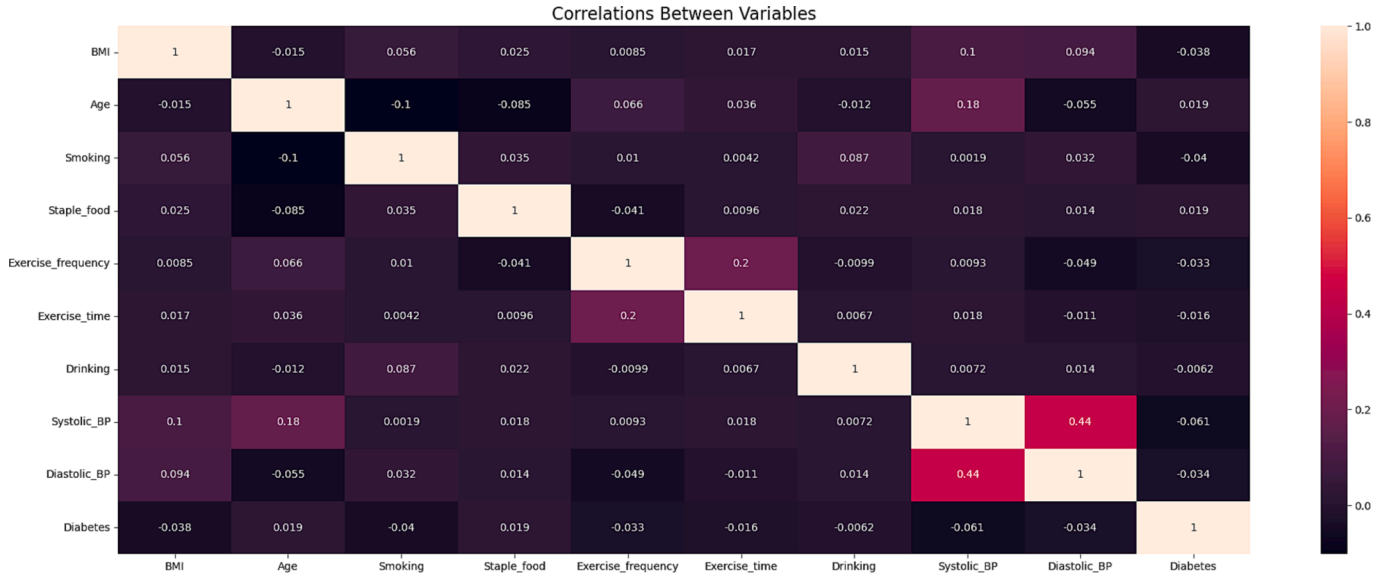



**Fig. 4.** Correlation coefficient matrix of 9 kinds of characteristic variables.

### 2.2. Patients and data sets

We obtained follow-up record data of people with diabetes from 2016 to 2023 from the grassroots community service management information system in Haizhu District, Guangzhou. In order to eliminate the noise in the initial data and obtain high-quality data, we have done a series of data preprocessing.

First, the text feature code is converted into a discrete variable, and the abnormal range data is eliminated. Then, the samples with missing values are deleted, and the duplicate samples are deleted. The final processed data set has a total of 252,176 records, including 9 features. (Table 1 and Fig. 2). The age range of the people with diabetes was 50–75 years, and the age distribution of healthy and people with diabetes was almost comparable (Fig. 3).

### 2.3. Characteristic selection of diabetes

Feature selection is a crucial step in building a predictive model, which helps to identify the most relevant features and reduce the dimensionality of the dataset. We evaluated three feature selection methods, comparing the relationship between different numbers of features and the AUC score of a random forest classifier (Pseudocode see Appendix A). The first method is SelectKBest (Naidu et al., 2021) using the f_classif evaluation function, which is very effective in selecting the K most relevant features. The second approach is to use the Recursive Feature Elimination (RFE) algorithm (Darst et al., 2018), which ranks features according to their importance and recursively selects the best subset of features. A third approach is to use rank ordering of feature importance in a random forest classifier (Behnamian et al., 2017). Detailed description see Appendix B.

### 2.4. Classification algorithm

The random forest classifier was used as the base classifier to generate the best features. After sorting the features using the above feature selection techniques, the predictive performance of the four machine learning methods (Dritsas and Trigka, 2022; Kumar, 2021; Xu et al., 2017; Xiong et al., 2019; Xie et al., 2019; Fitriyani et al., 2019) was evaluated (Pseudocode see Appendix C), respectively It is Random forest (RF) (Yang et al., 2023); eXtreme Gradient Boosting (XGB) (Yong-sheng et al., 2023; Zhijie et al., 2023; Linlin et al., 2023), K-Nearest Neighbors (KNN) (Deng Li et al., 2022), Ensemble Learning (VC) to achieve the best prediction. Detailed description see Appendix D.

### 2.5. Model evaluation

We evaluate model results using a confusion matrix (Jun Li et al., 2021; Fazakis et al., 2021). Four evaluation indicators are used: accuracy (Acc), sensitivity (Sens), precision (Pre), and F1 score.

(1) Accuracy (Acc) is the percentage of correct predictions made by the classifier in the testing phase compared to the actual value of the target.

$$Acc = \frac{(TP + TN)}{(TP + TN + FP + FN)} * 100\% \quad (1)$$

(2) Sensitivity (Sens) provides information on the percentage of true positives that were correctly classified during testing.

$$Sens = \frac{TP}{(TP + FN)} * 100\% \quad (2)$$

(3) Specificity (Spec) provides information on true negatives that were correctly classified during testing.

$$Spec = \frac{TN}{(TN + FP)} * 100\% \quad (3)$$

(4) Precision (Pre) is the percentage of instances marked as positive by the classifier relative to the total predicted positives (the accuracy of the classifier).

$$Pre = \frac{TP}{(TP + FP)} * 100\% \quad (4)$$

(5) The F1 score shows the harmonic mean of precision and recall.

$$F1score = \frac{2*TP}{(2*TP + FN + FP)} * 100\% \quad (5)$$

### 2.6. Diabetes risk score card

In order to promote clinical application more intuitively, we designed a random forest-based scorecard to assess the risk of diabetes. The scorecard uses the best feature subset obtained in feature selection, and realizes the segmentation of variables by binning the variables. After binning, the variables can use the standard scorecard format, that is, to score different segments. After binning, we get a series of discrete variables. We use the supervised encoding method WOE (weight of

**Table 2**
Feature Importance Coefficient Results using Three Feature Selection Techniques (Unit: %).

| SelectKBest | | RFE | | MDI | |
|---|---|---|---|---|---|
| **FEATURE** | **Importance coefficient** | **FEATURE** | **Importance coefficient** | **FEATURE** | **Importance coefficient** |
| BMI | 26.32 | BMI | 26.20 | BMI | 26.20 |
| Age | 18.0.66 | Age | 18.58 | Age | 18.58 |
| Systolic_BP | 1612 | Systolic_BP | 16.31 | Exercise_frequency | 16.31 |
| Diastolic_BP | 15.32 | Diastolic_BP | 15.61 | Drinking | 15.61 |
| Staple_food | 11.55 | Staple_food | 11.17 | Systolic_BP | 11.17 |
| Exercise_frequency | 5.28 | Exercise_time | 5.17 | Staple_food | 5.17 |
| Exercise_time | 4.98 | Exercise_frequency | 5.13 | Diastolic_BP | 5.13 |
| Smoking | 1.76 | Drinking | 1.82 | Exercise_time | 1.82 |
| Accuracy | 91.26 | Accuracy | 91.28 | Accuracy | 91.28 |
| F1 score | 91.73 | F1 score | 91.75 | F1 score | 91.75 |

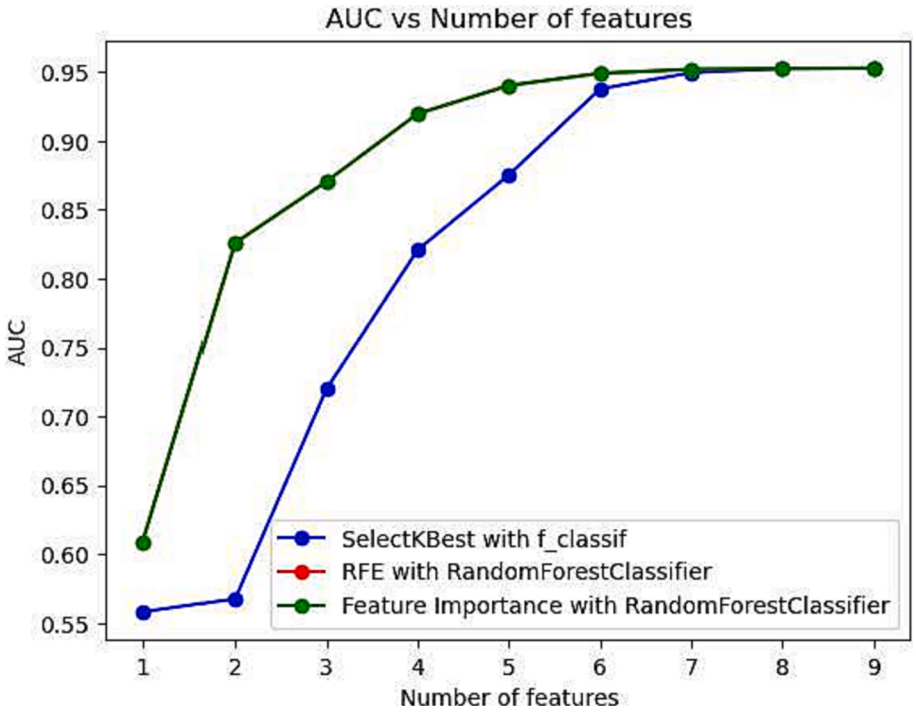



**Fig. 5.** Feature Number Results using Three Feature Selection Techniques.

evidence) to encode the variables and convert the discrete variables into continuous variables. We use univariate screening based on the IV value to remove features that are not very relevant to the target variable, eliminate variables due to linear correlation, and avoid feature redundancy. And use confusion matrix, TPR (Recal), FPR, ROC, KS value to evaluate the model. Finally, the scorecard was calibrated to determine a total score between 0 and 100 points. Detailed description see Appendix E.

### *2.7. Statement*

Guangzhou Haizhu District Community Health Development Guidance Center authorized the data used in our study, and waived the requirement of informed consent. All methods were carried out in accordance with relevant regulations.

## 3. Results

### *3.1. Characteristic selection*

The dataset we adopted contains a total of 9 clinical features. For redundant feature detection, we first use the Pearson correlation coefficient to reveal the correlation between any two features, and then, we use a heatmap to visualize the correlation (Fig. 4). As shown in the figure, Systolic_BP is positively correlated with Diastolic_BP, Exercise_-frequency is also positively correlated with Exercise_time, and Staple_-food is negatively correlated with Age. Therefore, feature selection should be used to exclude these redundant features.

We use three feature selection methods, and we compare the relationship between different numbers of features and the AUC score of a random forest classifier. The first method is SelectKBest using the f_classif evaluation function, which is very effective in selecting the K most relevant features. The second approach is to use the Recursive Feature Elimination (RFE) algorithm, which ranks features according to their importance and recursively selects the best subset of features. A third approach is to use rank ordering of feature importance in a random forest classifier. We split the dataset into training and testing sets and performed 5-fold cross-validation on the training set to evaluate the performance of each feature selection method using the area under the ROC curve (AUC) as a performance metric. We varied the number of features selected from 1 to 9 and plotted AUC vs. number of features for each method, finding that all methods had the best AUC score when using 8 features. The detailed results are shown in Table 2 and Fig. 5, and the corresponding confusion matrix and ROC curve are shown in

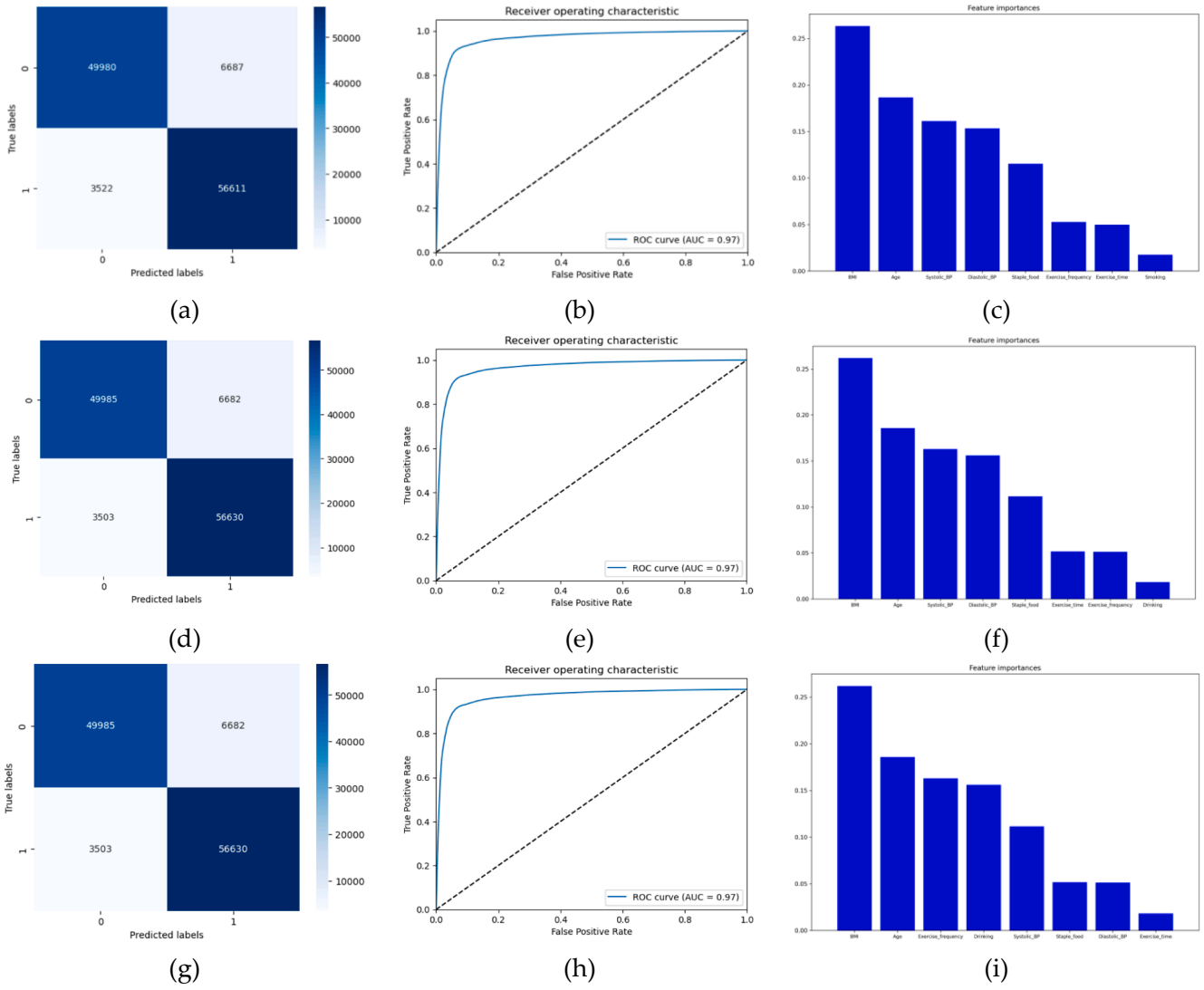



**Fig. 6.** Confusion matrices and ROC curves using three feature selection techniques. (a) The confusion matrix using the SelectKBest method; (b) ROC curve using the SelectKBest method; (c) The importance of features using the SelectKBest method; (d) The confusion matrix using the RFE method; (e) ROC curve using the RFE method; (f) The importance of features using the RFE method; (g) The confusion matrix using the MDI method; (h) ROC curve using the MDI method; (i) The importance of features using the MDI method.

Fig. 6.

We note that the curves perfectly coincide when using Recursive Feature Elimination (RFE) and ranking feature importance in the Random Forest classifier. The optimal features (BMI, Age, Systolic_BP, Diastolic_BP, Staple_food, Exercise_frequency, Exercise_time, Smoking) produced by SelectKBest using the f_classif evaluation function can achieve 91.26% accuracy; the 8 optimal features produced using recursive feature elimination (RFE) (BMI, Age, Systolic_BP, Diastolic_BP, Staple_food, Exercise_time, Exercise_frequency, Drinking) can achieve 91.28% accuracy; use the optimal features (BMI, Age, Exercise_frequency, Drinking, Systolic_BP, Staple_food, Diastolic_BP, Exercise_time) can achieve 91.28% accuracy. The performance of the three feature selection techniques is almost the same when selecting the 8 optimal features, all of which can obtain an AUC performance of 0.97. The performance of SelectKBest and the other two methods is similar, while the method using feature importance performs slightly worse, especially when the number of features is small. The optimal number of features is different for different methods, using the RFE method and random forest classifier requires the least number of features to achieve the highest AUC. Regardless of the feature selection technique used, the best feature subset always contained 7 features (BMI, Age, Systolic_BP, Diastolic_BP, Staple_food, Exercise_frequency, Exercise_time), indicating the importance of these features in diabetes diagnosis. Adding more features to the optimal feature subset does not improve the performance of the predictive model, which also indicates that these features are redundant information or noise.

**Table 3**
Performance metrics comparison for machine learning models in diabetes prediction(Unit: %).

| Methods | Accuracy | Precision | Recall | F1_score | ROC |
|---|---|---|---|---|---|
| RF | 91.24 | 89.36 | 94.22 | 91.72 | 91.15 |
| XGB | 67.98 | 67.67 | 72.39 | 69.95 | 67.85 |
| KNN | 74.91 | 72.24 | 83.27 | 77.36 | 74.66 |
| VC | 85.56 | 82.08 | 92.05 | 86.78 | 85.36 |

### 3.2. Risk assessment of diabetes

We use random oversampling RandomOverSampler to randomly sample from under-sample categories and add the sampled samples to the dataset to balance the dataset (Perveen et al., 2019). Using the 8 best

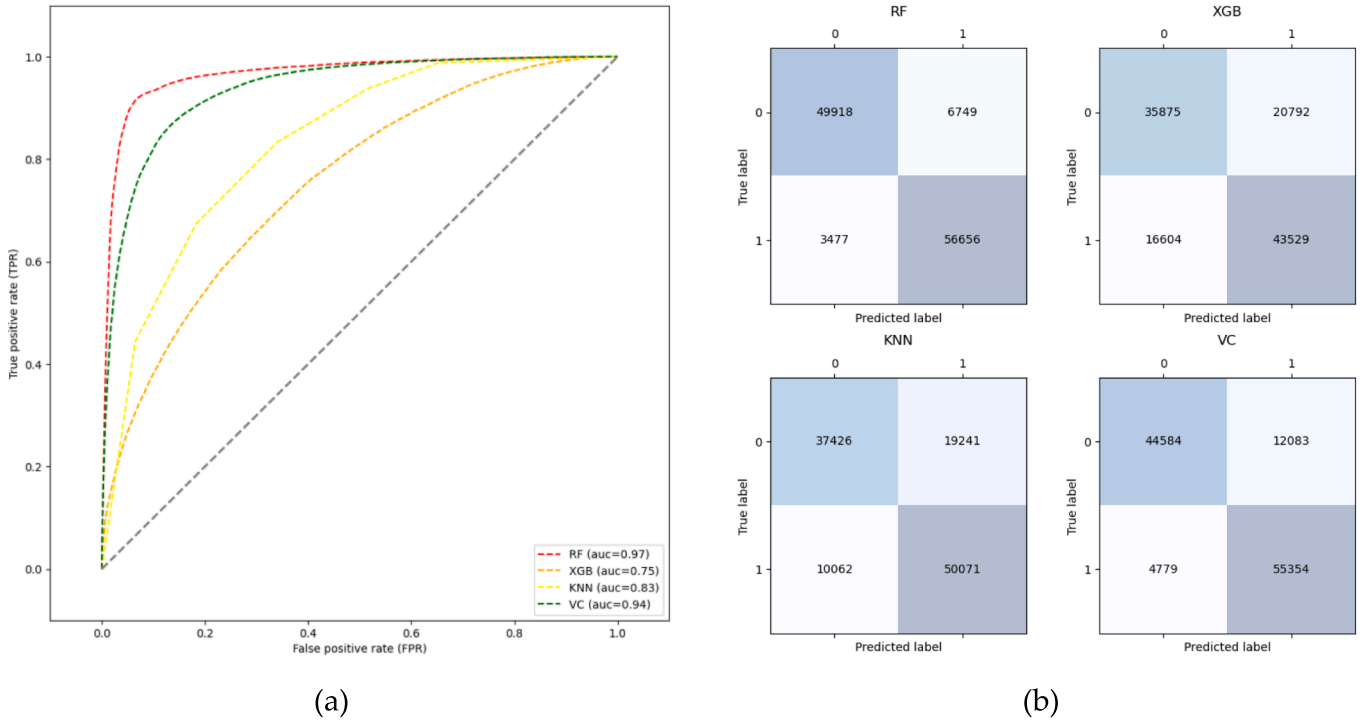



**Fig. 7.** ROC curves and confusion matrices for diabetes risk assessment using RF, XGBoost, KNN, and ensemble learning. **(a)** ROC curves for diabetes risk assessment using RF, XGBoost, KNN, and ensemble learning; **(b)** Confusion matrices for diabetes risk assessment using RF, XGBoost, KNN, and ensemble learning.

**Table 4**
Diabetes Risk Score Card.

| Base_score:50 | | | | | |
|---|---|---|---|---|---|
| Score boundary: [49.70, 50.71] | | | | | |
| **Feature** | **Threshold** | **Score** | **Feature** | **Threshold** | **Score** |
| BMI | (-inf, 21.79] | −0.41 | Age | (-inf, 55.0] | 0.08 |
| | (21.79, 23.02] | −0.27 | | (55.0, 61.0] | 0.15 |
| | (23.02, 24.24] | 0.11 | | (61.0, 69.0] | −0.10 |
| | (24.24, 26.04] | 0.02 | | (69.0, 77.0] | 0.09 |
| | (26.04, inf] | 0.60 | | (77.0, inf] | −0.25 |
| Staple_food | (-inf, 250.0] | 0.09 | Systolic_BP | (-inf, 120.0] | −0.09 |
| | (250.0, 300.0] | 0.18 | | (120.0, 126.0] | −0.12 |
| | (300.0, 330.0] | −0.00 | | (126.0, 130.0] | −0.06 |
| | (330.0, 400.0] | 0.03 | | (130.0, 134.0] | 0.03 |
| | (400.0, inf] | −0.51 | | (134.0, inf] | 0.37 |
| Exercise_frequency | (-inf, 3.0] | −0.19 | Diastolic_BP | (-inf, 72.0] | −0.06 |
| | (3.0, 4.0] | 0.11 | | (72.0, 75.0] | −0.09 |
| | (4.0, 5.0] | 0.07 | | (75.0, 78.0] | −0.05 |
| | (5.0, inf] | 0.59 | | (78.0, 82.0] | 0.02 |
| | | | | (82.0, inf] | 0.24 |
| Exercise_time | (-inf, 30.0] | −0.07 | | | |
| | (30.0, 45.0] | 0.26 | | | |
| | (45.0, inf] | 0.16 | | | |

features selected by RFE, we compared the performance of four different classifier models (RF, XGB, KNN, and VC) and calculated the evaluation metrics (Accuracy, Precision, Recall, F1 score and ROC), as shown in Table 3 and Fig. 7. It can be seen that the random forest-based classifier can achieve 91.24% accuracy, 89.36% precision, 94.22% recall rate, 91.72% F1 score, and 91.15% ROC performance. All indicators are higher than the other three Therefore, the random forest classifier was chosen as the final algorithm for constructing the diabetes risk assessment model.

### 3.3. Diabetes prediction risk assessment card

Community follow-up work is the daily work of community doctors to carry out early screening and management of chronic diseases of diabetes. Through mining the big data of community follow-up, the establishment of diabetes risk score card (Fan and Ding, 2022; Wang et al., 2021) can help community doctors to assess the risk of diabetes by calculating community residents' scores through important life characteristics. According to the definition of diabetes (the fasting blood glucose of normal people is lower than 6.1 mmol/L, and the fasting blood glucose is greater than or equal to 7.0 mmol/L, which meets the diagnostic criteria of diabetes), we made classified statistics on the fasting blood glucose of the data set according to "<6.1″, ">=6.1,<7.1″, ">=7.1″. "<6.1″ accounts for 30%, ">=6.1,<7.1″ accounts for 56%, and ">=7.1″ accounts for 14%. Therefore, we set two scoring points, 30 and 86. The total score can be divided into three intervals, corresponding to

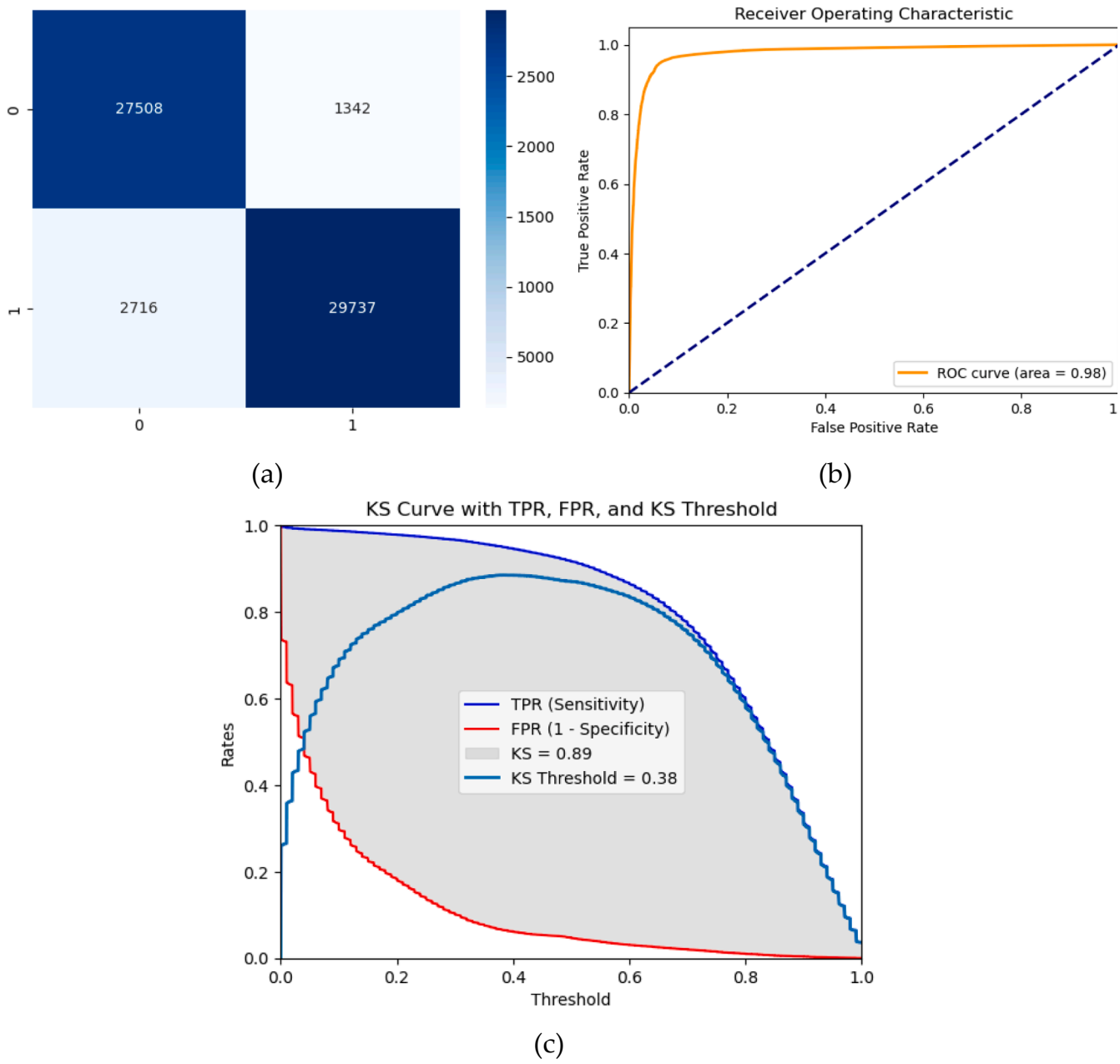



**Fig. 8.** Establishment of the Diabetes Risk Assessment Score Model. (a) Confusion matrix for validating the performance of the credit scoring model; **(b)** ROC curve for validating the performance of the credit scoring model; **(c)** TPR, FPR, and KS curve for validating the performance of the credit scoring model.

low, medium, and high risk levels. The main characteristic box splitting results are shown in Table 4.

To verify the validity of the score card, we store the algorithm model and use the original data to make regression prediction through the model. The confusion matrix, ROC, TPR, FPR, KS curves obtained are shown in Fig. 8. The accuracy is 93.38%, the Precision is 95.68%, the Recall is 91.63%, and the F1 Score is 93.61%. It can be seen that compared with the diabetes risk assessment model, the performance loss of the score card is quite small.

We made statistics on the original fasting blood glucose values corresponding to the samples in each section of the three sections of the evaluation card in the test results. The average fasting blood glucose values in each section of the Score boundary: (- inf, 49.70] is 5.59, (49.70, 50.71] is 6.58, (50.71, inf] is 9.29. It is consistent with the clinical definition of diabetes risk interval ("<6.1″, ">=6.1,<7.1″, and ">=7.1″). It shows that the method of establishing diabetes risk score card is reasonable.

## 4. Discussion

### 4.1. Comparison with prior work

To demonstrate the performance of the diabetes prediction model in this study, we compared it with other research results using the same performance metrics, as shown in Table 5. The dataset used in this study has a large sample size, and the model exhibits a favorable advantage in terms of accuracy, with a more balanced performance. The random forest algorithm shows notable advantages in handling risk prediction models.

**Table 5**
Model information and performance comparisons for related research work.

| Authors | Optimal algorithm | Used database | Accuracy (%) | Recall (%) | AUC (%) |
|---|---|---|---|---|---|
| Haohui Lu et al. | Random forest | Administrative healthcare claim data | 84.95 | 84.95 | 91.09 |
| Hang Lai et al. | Gradient BoostingMachine | The most recent records of 13,309 Canadian patients aged between 18 and 90 years | \ | 71.6 | 84.7 |
| Siyu Liu et al. | Back propagation | Sample Data of Residents in Jilin Province, China | 91.2 | \ | 78 |
| Salliah Shafi Bhat et al. | Random forest | A clinical dataset in the Indian district of Bandipora in the years April 2021–Feb2022 | 98 | 87.5 | 67.2 |
| Our work | Random forest | Follow-up records of diabetic patients in Haizhu District, Guangzhou, from 2016 to 2023. | 91.24 | 94.22 | 91.15 |

**Table 6**
Performance comparison of 4 algorithms after removing important features.

| | | Accuracy (%) | Precision (%) | Recall (%) | F1_score (%) | ROC (%) |
|---|---|---|---|---|---|---|
| Delete BMI | RF | 86.69 | 83.98 | 91.63 | 87.64 | 86.54 |
| | XGB | 65.05 | 64.96 | 69.71 | 67.25 | 64.90 |
| | KNN | 71.95 | 69.70 | 80.52 | 74.72 | 71.69 |
| | VC | 81.87 | 78.46 | 89.31 | 83.53 | 81.65 |
| Delete Age | RF | 88.69 | 85.93 | 93.31 | 89.47 | 88.55 |
| | XGB | 65.90 | 65.26 | 72.20 | 68.55 | 65.71 |
| | KNN | 71.68 | 69.24 | 80.95 | 74.64 | 71.39 |
| | VC | 82.89 | 79.13 | 90.70 | 84.52 | 82.66 |
| Delete BMI & Age | RF | 74.28 | 71.92 | 82.09 | 76.67 | 74.04 |
| | XGB | 62.46 | 62.12 | 69.41 | 65.56 | 62.25 |
| | KNN | 65.94 | 65.09 | 72.99 | 68.81 | 65.72 |
| | VC | 71.90 | 69.95 | 79.64 | 74.48 | 71.66 |

### *4.2. Feature importance verification*

Regardless of the feature selection technique used, the optimal feature subset always contained 7 features: BMI, Age, Systolic_BP, Diastolic_BP, Staple_food, Exercise_frequency, Exercise_time, indicating the importance of these features in diabetes diagnosis. After we tried to remove the BMI feature with the highest importance coefficient on the basis of 3.2, the accuracy rate of the random forest model was 86.69%, and the accuracy rate after removing the age feature was 88.69%. After removing these two features at the same time, the accuracy rate dropped significantly, which was 74.28%. The situation of removing these two features by various algorithms is shown in Table 6. We further statistically analyzed the distribution of these 7 important features to present the differences between people with diabetes and healthy individuals in detail (Fig. 9). It can be seen that except for Staple_food and Exercise_time, the other five indicators show statistical differences, which help to predict the risk of diabetes. We also tried to remove Staple_food, Exercise_time, and Smoking on the basis of 3.2, and the model accuracy was 89.14%. It can be seen that compared with the accuracy of 91.24% of the optimal features in Table 2, the performance of the model drops significantly after removing the key features (see confusion matrix in Fig. 10). Fusion can improve the robustness, reliability and accuracy of the proposed diabetes diagnosis model.

Through the statistical analysis of the distribution map, it can be seen that the individual's BMI, age, and systolic blood pressure are closely related to diabetes and are the most important risk factors. The individual's disease risk increases with age, increases with BMI, increases with systolic blood pressure. Therefore, based on the important characteristics and risk factors, we should strengthen community diabetes prevention work, improve people's health awareness, change unhealthy lifestyles, improve personal lifestyles through self-examination, and reduce the level of risk factors.

Community doctors are the gatekeepers of basic level health security. We can use our model to conduct large-scale early diabetes screening. If diabetes patients are found early, the early diagnosis and early treatment rate of diabetes can be improved. Identifying people with high levels of diabetes and conducting health guidance and intervention against their risk factors can reduce or delay the occurrence of diabetes. The analysis data we use directly comes from the information system of the daily work of the grass-roots community, without secondary processing. Therefore, our diabetes risk assessment model and score card can be directly applied to the grass-roots community system, for large-scale risk identification and early warning of community doctors, to further promote diabetes prevention and control strategies, which has important social significance for the early control of diabetes.

## 5. Conclusions

In our study, we designed a diabetes risk assessment model based on random forest classifier and a diabetes risk assessment card based on the follow-up record data of diabetes patients obtained from the grassroots community service management information system in Haizhu District, Guangzhou, to improve the applicability of the model in clinical and real life. The risk assessment card can calculate the risk of diabetes according to eight optimal characteristic parameters. After the assessment of the model, the model has high credibility. The results can be used for community doctors to screen and early warn diabetes on a large scale, and can also be used for wearable devices or intelligent biosensors for individual patient self examination, in order to improve lifestyle and reduce risk factor levels.

This study also has the limitation that the characteristics are not comprehensive enough, which may affect the accuracy of disease risk prediction to a certain extent. However, it is still feasible to train and predict more and more comprehensive feature inputs according to the method proposed in the study and the prediction model used, and it can further improve the accuracy of the prediction results of the model. In further research, we will extract more and more comprehensive characteristic variables for research, we will consider conducting time series analysis of follow-up data and consider time series analysis of follow-up data to analyze related variables that affect blood sugar improvement, so as to better guide the self-health management of the sick population.

## Author contributions

Liangjun Jiang conceived and designed this study, and wrote the manuscript. Zhenhua Xia conducted code design. Haimei Gong and Jing Wang assisted in literature search. Ronghui Zhu and Juan Li explained the data. Lei Wang conducted research supervision and made key revisions to the manuscript. All authors have read and agreed to the published version of the manuscript.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

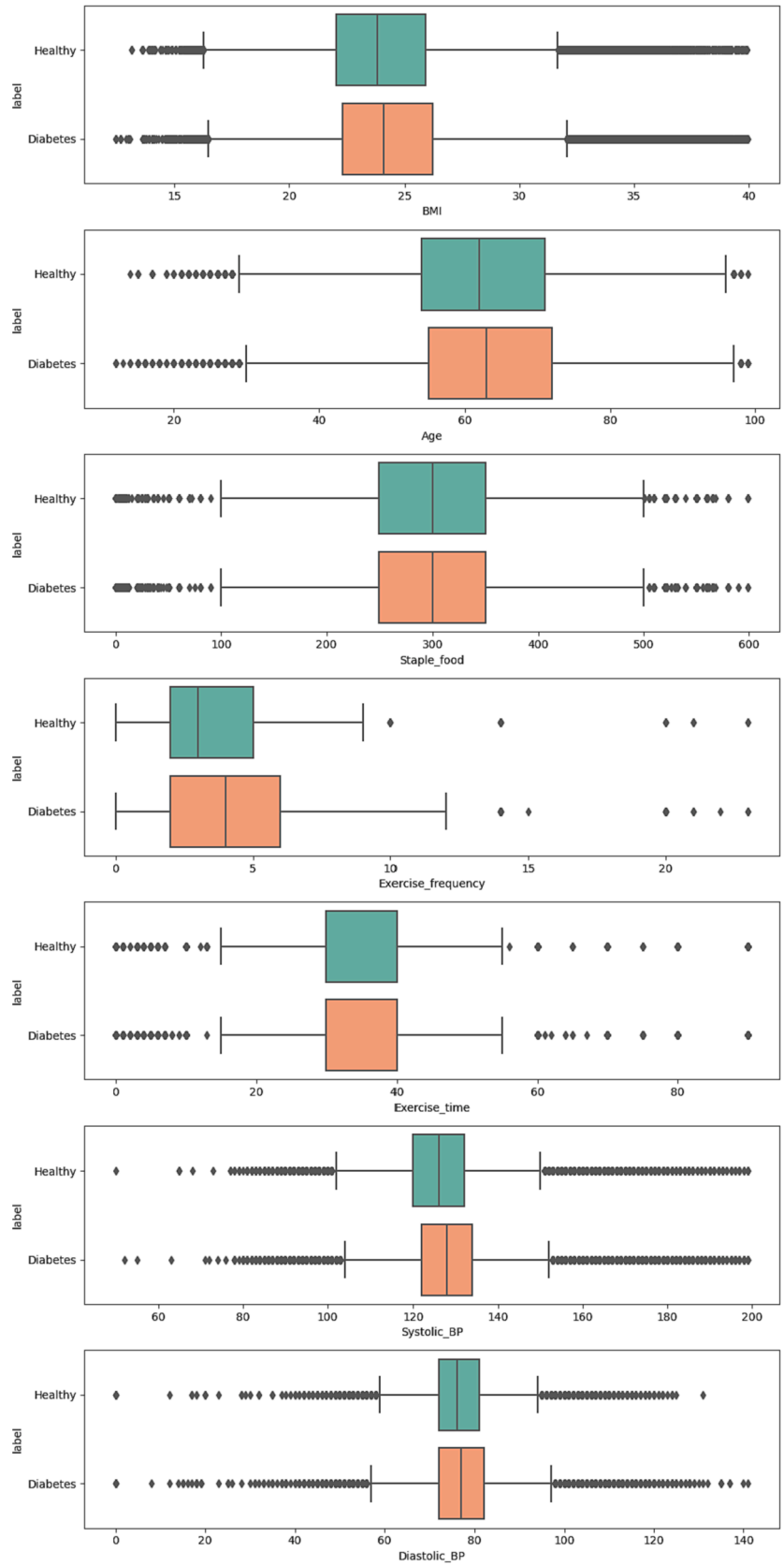



**Fig. 9.** Statistical analysis of the seven best features in healthy individuals and people with diabetes.

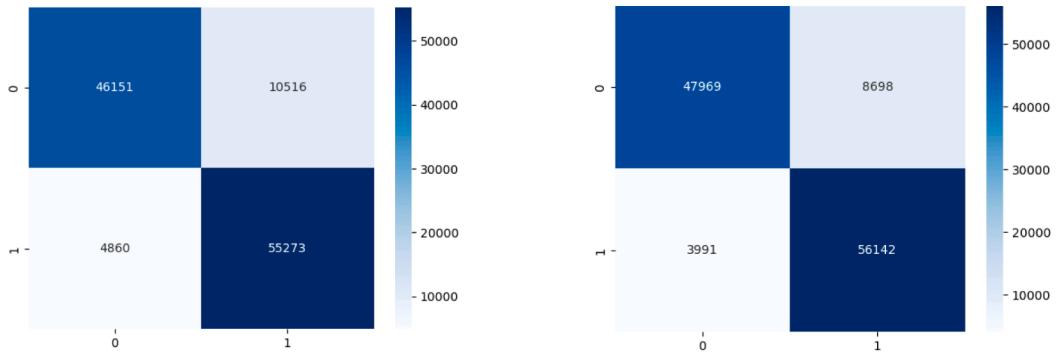



**Fig. 10.** Establishment of the Diabetes Risk Identification Card Model. **(a)** Confusion matrix of the model after removing the BMI feature with the highest importance coefficient; **(b)** Confusion matrix of the model after removing the Staple_food, Exercise_time, and Smoking features.

## Data availability

The data presented in this study are available on request from the corresponding author. The data are not publicly available due to privacy.

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## Appendix A

**Require:** $dataset$, $max_k$
$X, y \leftarrow$ split $dataset$ into features $X$ and labels $y$
$X_train, X_test, y_train, y_test \leftarrow$ split $X$ and $y$ into training and testing sets with a test size of 0.3
$aucs_kbest \leftarrow []$
**for** $k$ in range(1, $max_k$+1) **do**
$kbest \leftarrow$ SelectKBest(f$_classif, k$ =k)X$_train_kbest \leftarrow$ fit $kbest$ on $X_train$ and transform $X_train$
$clf \leftarrow$ RandomForestClassifier()
$scores \leftarrow$ cross-validation scores of $clf$ on $X_train_kbest$ and $y_train$ with 5 folds using 'roc$_auc' scoring append the mean of scores to aucs_kbest$
**end for**
$aucs_rfe \leftarrow []$
**for** $k$ in range(1, $max_k$+1) **do**
$rfe \leftarrow$ RFE(estimator=RandomForestClassifier(), n$_features_to_select$ =k)X$_train_rfe \leftarrow$ fit $rfe$ on $X_train$ and transform $X_train$
$clf \leftarrow$ RandomForestClassifier()
$scores \leftarrow$ cross-validation scores of $clf$ on $X_train_rfe$ and $y_train$ with 5 folds using 'roc$_auc' scoring append the mean of scores to aucs_rfe$
**end for**
$aucs_fi \leftarrow []$
**for** $k$ in range(1, $max_k$+1) **do**
$fi \leftarrow$ RandomForestClassifier(max$_features$ =k)scores $\leftarrow$ cross-validation scores of $fi$ on $X_train$ and $y_train$ with 5 folds using 'roc$_auc' scoring$
append the mean of $scores$ to $aucs_fi$
**end for**
plot $aucs_kbest$ against the range 1 to $max_k$ with blue color and label 'SelectKBest'
plot $aucs_rfe$ against the range 1 to $max_k$ with green color and label 'RFE'
plot $aucs_fi$ against the range 1 to $max_k$ with red color and label 'Feature Importance'
set title of the plot as 'Comparison of Feature Selection Methods'
set x-axis label as 'Number of Features'
set y-axis label as 'AUC'
add legend to the plot
display the plot

**Algorithm 1. Pseudocode for f**eature Selection Comparison.

## Appendix B

### *B.1. SelectKBest*

SelectKBest is a feature selection method based on statistical principles. The principle is to calculate the correlation between each feature and the target variable, and then select the top K features most correlated with the target variable. This process can be achieved by the following steps:

(1) Calculate the correlation between each feature and the target variable. This can be achieved by calculating Pearson correlation coefficient, chi-square test, mutual information and other methods.
(2) Calculate the score of each feature. The higher the score, the stronger the correlation between the feature and the target variable.
(3) The top K features with the highest scores are selected as the best features.

### *B.2. Recursive feature elimination*

The main idea of recursive feature elimination is to repeatedly build the model, then select the best (or worst) features (based on the coefficients), put the selected features aside, and repeat the process for the remaining features until all features have been traversed. The order of elimination in this process is to sort the features.

### *B.3. Feature importance ranking in the random forest classifier*

The importance of a feature X in a random forest is calculated as follows:

(1) For each decision tree in the random forest, use the corresponding OOB (out of pocket data) data to calculate its out of pocket data error, which is recorded as errOOB1.
(2) Randomly add noise interference to the feature X of all samples in the out of bag data OOB (the value of the sample at feature X can be randomly changed), and calculate its out of bag data error again, expressed as errOOB2.
(3) Assuming that there are N trees in the random forest.

$$\text{The importance of feature X} = \sum(\text{errOOB2} - \text{errOOB1})/\text{Ntree}. \tag{1}$$

The reason why this expression can be used as a measure of the importance of the corresponding feature is that if after adding noise randomly, the out-of-bag accuracy of the feature is greatly reduced, which means that this feature has a great influence on the classification result of the sample, that is, its importance is relatively high.

## Appendix C

```
Require:
    data_file_path: path to the CSV file containing the data
    label_col: name of the column containing the labels
Procedure ensemble_classifier (data_file_path, label_col)
    data ← read CSV file from data_file_path and drop rows with missing values
    feature_cols← ['BMI', 'Sex ID', 'Age', 'Smoking', 'Staple food', 'Exercise frequency', 'Exercise time', 'Drinking', 'Systolic BP', 'Diastolic BP']
    data[label_col] ← convert the label column to numerical values
    X ← features from data using feature_cols
    y ← labels from data
    correlations ← compute correlation matrix between features in X
    k ← 8
    sorted_columns← top k correlated feature columns in X based on correlation matrix
    X_new← features from X using sorted_columns and convert to array
    X_train, X_test, y_train, y_test ← split data into training and testing sets
    class_weight ← 0: mean of 0 in y_train, 1: mean of 1in y_train
    scaler ← initialize a StandardScaler
    X_train ← standardize training features using scaler
    X_test ← standardize testing features using scaler
    model1 ← initialize a RandomForestClassifier with 100 estimators and random state of 42
    model2 ← initialize an XGBClassifier with random state of 42
    model3 ← initialize a KNeighborsClassifier
    VC ← initialize a VotingClassifier with model1, model2, model3 and soft voting
    models ← [model1, model2, model3, VC]
    for model in models do
        model.fit(X_train, y_train)
        training_accuracy ← accuracy of model on training set
        testinga_ccuracy ← accuracy of model on testing set
        print type(model).name: Training_accuracy = training_accuracy,
                                Testinga_ccuracy = testinga_ccuracy
    end for
end procedure
```

Algorithm 2: **Pseudocode for e**nsemble Classifier.

## Appendix D

### *D.1. Random Forest (RF)*

Random forest is an integrated learning algorithm based on decision trees, which determines the final prediction result through the voting results of multiple decision trees. For the diabetes prediction problem, random forest can use multiple decision trees to learn the relevant characteristics and potential laws of diabetes, so as to predict the probability of disease. In each decision tree, random forest randomly selects a part of features for training, thus avoiding the phenomenon of overfitting.

Random forest uses tree classifier $\{h(x, \phi_q), q = 1, 2, ..., Q\}$ to simulate data samples. Input sample data into the data processing system, determine the data direction through the vector $h(x, \phi_q)$ of the classification regression tree, and select the final classification result through simple and multiple filtering rules. The formula is:

$$H(x) = max\sum_{q=1}^{Q} i[h(x, \phi_q) = I] \tag{2}$$

In the formula, Q is the total number of branches of the tree classifier; $I$ represents the target sample data for classification direction.

In random forest, Gini index $Gini(N)$ is selected to determine the characteristic node to construct the data decision tree:

$$Gini(N) = 1 - \sum_{k=1}^{K2}(|N_k|/N) \tag{3}$$

$N_k$ represents a feature k randomly selected from the data feature sample set $N$.

### *D.2. eXtreme Gradient Boosting (XGB)*

XGBoost is an integrated learning algorithm based on gradient boosting trees, which improves prediction accuracy by iteratively training multiple decision trees. In each iteration, XGBoost will adjust the weight of the next training sample according to the error between the previous prediction result and the actual label, so that the model can gradually learn the relevant characteristics and potential laws of diabetes. At the same time, XGBoost also uses regularization technology to avoid overfitting.

XGBoost is defined as:

$$\widehat{y}_i^m = \sum_{z=1}^{Z} f_z(x_i) = \widehat{y}_i^{(m-1)} + f_t(x_m) \tag{4}$$

In the formula, $\widehat{y}_i^m$ is the prediction result of sample i after m iterations, $Z$ is the set of all CART trees, $f_z$ is the prediction result of the z-th tree, $x_i$ is the i-th input sample, $\widehat{y}_i^{(m-1)}$ is the prediction result of the previous m-1 tree, and $f_t(x_m)$ is the prediction result of the m-th tree.

The optimal objective function is:

$$Obj = -\frac{1}{2}\sum_{j=1}^{T}\frac{\left(\sum_{i\in I_j} g_i\right)^2}{\sum_{i\in I_j} h_i + \lambda} + \gamma T = -\frac{1}{2}\sum_{j=1}^{T}\frac{G_j^2}{H_j + \lambda} + \gamma T \tag{5}$$

In the formula, $G_j$ is the first-order degree statistics of the sample set of leaf node j, $G_j = \sum_{i\in I_j} g_i$; $H_j$ is the second-order gradient statistics of the leaf node j sample set, $H_j = \sum_{i\in I_j} h_i$.

### *D.3. K-Nearest Neighbors (KNN)*

KNN is an instance-based classification algorithm that determines the nearest K data points by calculating the distance between new data points and training data points, and then predicts the classification result by voting on the labels of these K data points. For the diabetes prediction problem, KNN can calculate the distance between different patients according to the input characteristics of the patient, such as BMI, blood pressure, etc., and predict the probability of disease according to the labels of the nearest K patients.

KNN needs to calculate the distance between each sample point and the test point, select the closest k samples, obtain their labels, find the label with the highest number among the k samples, and return the label. Let feature space X be n-dimensional real vector space $R^n$, $x_i, x_j \in X$, $x_i = \left(x_i^{(1)}, x_i^{(2)}, \ldots x_i^{(n)}\right)^T$, $x_j = \left(x_j^{(1)}, x_j^{(2)}, \ldots x_j^{(n)}\right)^T$, and $L_p$ distance of $x_i, x_j$ is defined as:

$$L_p(x_i, x_j) = \left(\sum_{l=1}^{n}\left|x_i^{(l)} - x_j^{(l)}\right|^p\right)^{\frac{1}{p}} \tag{6}$$

We use Euclidean distance measurement to identify the k closest training samples in the training set, select the category with the most occurrences of these k samples, and label them as the prediction result.

$$L_2(x_i, x_j) = \left(\sum_{l=1}^{n}\left|x_i^{(l)} - x_j^{(l)}\right|^2\right)^{\frac{1}{2}} \tag{7}$$

By using the distance between samples as a non similarity indicator between samples and making decisions based on the dominant category among k objects, the accuracy of the algorithm is improved.

### *D.4. Ensemble Learning (VC)*

Ensemble Learning (VC) is the use of multiple models to train a dataset and ultimately use voting to select results. We trained the dataset using three models: Random forest (RF), eXtreme Gradient Boosting (XGB), and K-Nearest Neighbors (KNN), and used soft voting to select the results.

## Appendix E

### *E.1. Variable binning (Chimerge binning)*

The basic idea of Chimerge binning is to merge two adjacent intervals if they have similar class distributions, otherwise they should be kept separate. Chimerge usually uses the chi-squared value to measure the class distribution between two adjacent intervals. The specific algorithm of Chimerge is as follows:

Input: maximum number of intervals for binning.

Initialization:

(a) Continuous variables are sorted in ascending order, while discrete variables are first converted to the ratio of diabetes and then sorted in ascending order.
(b) In order to reduce computation, for variables with a state number greater than a certain threshold (1 0 0), coarse binning is performed using equi-depth binning.
(c) If there are missing values, the missing values are treated as a separate interval.

Merge intervals:

(a) Calculate the chi-squared value for each pair of adjacent intervals.
(b) Merge the pair of intervals with the smallest chi-squared value:

$$X^2 = \sum_{i=1}^{2}\sum_{j=1}^{2}\frac{\left(A_{ij} - E_{ij}\right)^2}{E_{ij}} \tag{8}$$

$A_{ij}$: The number of instances of class j in the i-th interval.

$E_{ij}$: $E_{ij} = \frac{N_i}{N} \times C_j$, $N$ represents the sample size of the merged interval, $N_i$ represents the sample size of the i-th group, and $C_j$ represents the number of

instances of the j-th class in the merged interval.

(c) Repeat the above two steps until the number of bins is no greater than n.

Post-processing after binning:

(a) Merge bins with a diabetes rate of 0 or 1 (a bin cannot consist entirely of diabetes patients or healthy people).
(b) Delete bins where the proportion of samples in a bin exceeds 95%.
(c) Check whether the proportion of people with diabetes with missing bins is equal to that of non-missing bins. If they are equal, merge them.

Output: binned data and binning intervals.

### *E.2. WOE Encoding*

WOE, short for weight of evidence, is a supervised encoding method that uses the concentration of the predicted class as the numerical value of the encoding. The WOE value is:

$$WOE_i = ln(\frac{p_{i_1}}{p_{i_0}}) = ln(\frac{\#B_i/\#B_T}{\#G_i/\#G_T}) \tag{9}$$

$p_{i_1}$ is the proportion of diabetes patients in the i-th box among all diabetes patients.
$p_{i_0}$ is the proportion of healthy individuals in the i-th box among all healthy individuals.
$\#B_i$ is the number of diabetes patients in the i-th box.
$\#G_i$ is the number of healthy individuals in the i-th box.
$\#B_T$ is the total number of diabetes patients.
$\#G_T$ is the total number of healthy individuals.
A simple transformation of the above formula yields:

$$WOE_i = ln(\frac{\#B_i/\#B_T}{\#G_i/\#G_T}) = ln(\frac{\#B_i/\#G_i}{\#B_T/\#G_T}) \tag{10}$$

The WOE can also be understood as the ratio of people with diabetes to healthy individuals in the current bin, and the difference between this ratio and the ratio of people with diabetes to healthy individuals in the entire sample (i.e., the random ratio). The larger the WOE, the greater the difference, and the higher the likelihood of people with diabetes in the current group. Conversely, the smaller the WOE, the smaller the difference, and the lower the likelihood of response in the current sample. When the ratio of people with diabetes to healthy individuals in the bin is equal to the random ratio, the bin has no predictive ability, i.e., WOE = 0.

### *E.3. Univariate screening based on IV value*

The formula for calculating the IV value of the ith bin in a variable is:

$$IV_i = (\frac{\#B_i}{\#B_T} - \frac{\#G_i}{\#G_T}) * ln(\frac{\#B_i/\#B_T}{\#G_i/\#G_T}) = (\frac{\#B_i}{\#B_T} - \frac{\#G_i}{\#G_T}) * WOE_i \tag{11}$$

The IV value of a variable is the sum of the IV values of all its corresponding bins:

$$IV = \sum_i^n IV_i \tag{12}$$

### *E.4. Convert to scorecard*

First, use random forest to train and predict the data, and obtain the probability of each sample belonging to each category. Then, based on the probability of each category, calculate the ratio of "diabetes patient probability/healthy population probability" for each sample. Next, use the ratio to calculate the score for each sample, using the same formula as the logistic regression scorecard:

$$\text{Score} = \text{A} - \text{B}*\ln(\text{diabetes patient probability}/\text{healthy population probability}) \tag{13}$$

Where A and B are constants that can be selected based on prior knowledge and experience. The probability of diabetes patients and the healthy population can be obtained from the random forest's predicted results. Finally, add up the scores for each sample and add the base score to get the final scorecard score.

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